

How to Build Your Pendant

From parts in boxes → a working pendant on a chain. Take your time. Each step is a single thing to do.

🕒 **Total time:** About 90 minutes per pendant the first time, 45 minutes once you've done one. Make 2 or 3 so you have spares.

✓ **Before you start:** Clear a desk. Have all parts in front of you. Put on a podcast. Take a breath. This is going to work.

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Put All Your Parts on the Desk

Take 2 minutes. If anything is missing, find it now, not mid-build.

The Brain & Power (already have these)

- ☐ **XIAO nRF52840 Sense board** — the tiny black-and-purple circuit board (you have 10)
- ☐ **500mAh drone LiPo battery** — the small silver pouch with red and black wires (or the EEMB 250mAh, either works)
- ☐ **USB-C cable** — short right-angle one
- ☐ **Silver chain** — from the 12-pack of 20" chains

The Tools (already have these)

- ☐ **Soldering iron** — TXINLEI 60W kit with the stand
- ☐ **Solder wire** — AUSTOR 60/40, 0.8mm
- ☐ **Flush cutters** — small black scissors-looking thing
- ☐ **Flux pen** — looks like a marker
- ☐ **Kapton tape** — orange thin tape on the roll
- ☐ **Foam mounting tape** — double-sided black tape

The Body (arriving Monday)

- ☐ **Endhokn white epoxy putty** — looks like a fat white candy bar in plastic
- ☐ **COORIART liquid chrome markers** — 3 silver pens, different tip sizes
- ☐ **Sculpting tools** — set of 27 small wooden/plastic tools
- ☐ **Hobby knife** — sharp X-acto style
- ☐ **Needle files** — set of 10 small files
- ☐ **Nitrile gloves** — for handling putty
- ☐ **Alcohol wipes** — for cleaning before chrome
- ☐ **Microfiber cloths** — for the brushed finish
- ☐ **Wooden toothpicks** — for the brushed finish
- ☐ **Jump rings + clasps kit** — small silver rings for the chain loop
- ☐ **Super glue** — backup adhesive

Build the Electronics

1

Solder the battery to the XIAO board. 20 minutes. The scary-sounding part is actually two tiny dots of metal.

1 Plug in the soldering iron and let it heat up

5 min

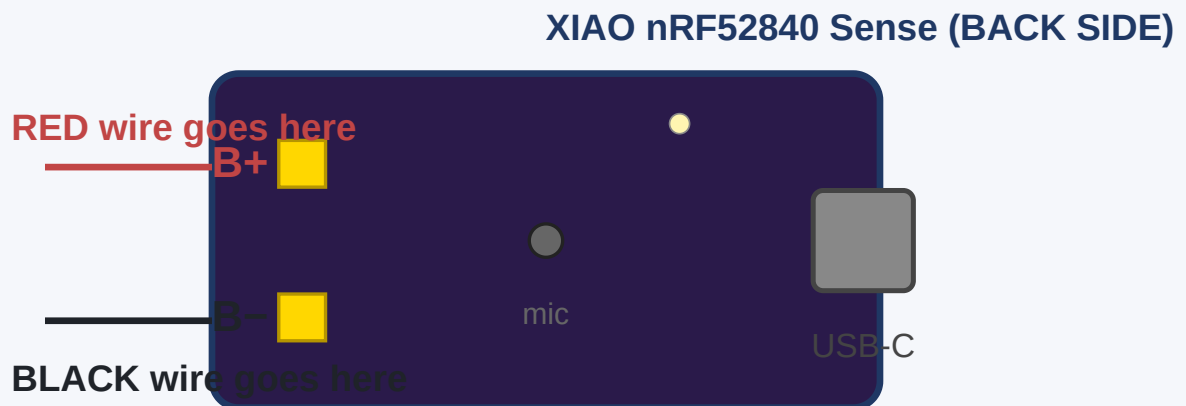
- 1 Stand the iron in its metal stand. Set it to **350°C / 660°F**.
- 2 Wait until the tip looks shiny silver. **Don't touch the metal part. Ever.** Only the rubber handle.
- 3 Wipe the tip on the wet sponge in the stand. It should sizzle.

 **SAFETY:** The tip is 350°C. It will burn through skin instantly. Put it back in the stand every single time you stop using it. Never leave it lying on the desk.

2 Look at your XIAO board, find the battery pads

2 min

- 1 Hold the XIAO board with the USB-C port pointing away from you.
- 2 Flip it over. On the back, near the USB-C end, you'll see two tiny silver squares labeled **B+** and **B-**.
- 3 **B+ is for the RED battery wire. B- is for the BLACK wire.** Don't mix these up.



Back side of XIAO board. Yellow squares are the battery pads.

3 Prepare the battery wires

3 min

- 1 Take a battery out of the bag. There's a red wire and a black wire. They might be too long — that's fine.
- 2 If the wire tips are not stripped (no silver showing), use the flush cutters to gently scrape off about **3mm** of the plastic at the end of each wire so the silver shows.
- 3 Twist the silver strands tight with your fingers so they don't fan out.

⚠ **Battery rule:** Never let the red and black wires touch each other. That's a short circuit. The battery will get hot and could pop. Keep them apart at all times.

4 Solder the red wire to the B+ pad

3 min

- 1 Use the flux pen to dab a small dot of liquid on the **B+** pad. This helps the solder stick.
- 2 Hold the RED battery wire on top of the B+ pad with one hand.
- 3 With the other hand, touch the iron tip to where the wire meets the pad. Count **"one, two"**.
- 4 Touch the solder wire to the iron tip. A small shiny blob will form. As soon as it covers the wire and pad, pull away the solder, then pull away the iron.
- 5 Don't move the wire for 5 seconds. Let it cool. The blob should look shiny, not dull.

✓ **Good solder joint:** Shiny silver dome covering the pad. Bad: dull gray, blobby, or a ball balanced on top. If it looks wrong, reheat it for 2 seconds and try again.

5 Solder the black wire to the B– pad

3 min

- 1 Same exact process as Step 4, but with the BLACK wire on the B– pad.
- 2 Make sure the red and black soldered joints are not touching each other.
- 3 Put the iron back in the stand. You're done soldering for today.

6 Test that it powers on

1 min

- 1 Plug the USB-C cable into the XIAO board (not into your Mac yet).
- 2 Plug the other end into your Mac.
- 3 Look at the board. **One or two tiny LEDs should glow.** If yes → the electronics work. Celebrate.

 **If nothing lights up:** Unplug immediately. Check that the red and black wires are NOT touching. Check that B+ has red and B- has black, not swapped. If still nothing, the board may have shorted — set it aside and use a new one. (You have 10.)

7 Stick the battery to the back of the board

2 min

- 1 Cut a small piece of the double-sided foam tape, about **15mm × 10mm**.
- 2 Peel one side. Stick it to the back of the battery (the side without the wires).
- 3 Peel the other side. Press the battery onto the back of the XIAO board, opposite the USB-C port. The wires should curve gracefully, not be stretched tight.
- 4 Wrap one piece of orange Kapton tape around the whole stack to keep it together.

 **You now have:** a working pendant brain. It's about 25mm × 22mm × 8mm. From now on, treat it like a tiny phone — don't drop it, don't get water on it.

2 Sculpt the Pendant Body

White epoxy putty around the electronics. Like wrapping a present in clay. 30 minutes per pendant.

8 Put on nitrile gloves

30 sec

- 1 Epoxy putty is sticky and a little irritating on bare hands. Always wear gloves while kneading.
- 2 Lay a piece of parchment paper or wax paper on your desk. Or use a clean plastic cutting board. Putty will not stick to either.

9 Cut a thumb-sized piece of putty

2 min

- 1 Open the putty tube. Use the hobby knife to slice off a piece about the size of your thumb knuckle. **≈ 20g**
- 2 Look at the slice. Endhokn putty has two colors swirled together (light and slightly darker white). You need to mix them.
- 3 Knead the piece in your hands for **2 minutes straight**, like you're warming up clay. Fold it, squish it, fold it again. It will turn one even color when fully mixed.



Clock starts NOW. Once mixed, you have about 5 minutes of working time before it starts setting. Move quick. Full cure takes 1 hour.

10 Flatten it into a pancake

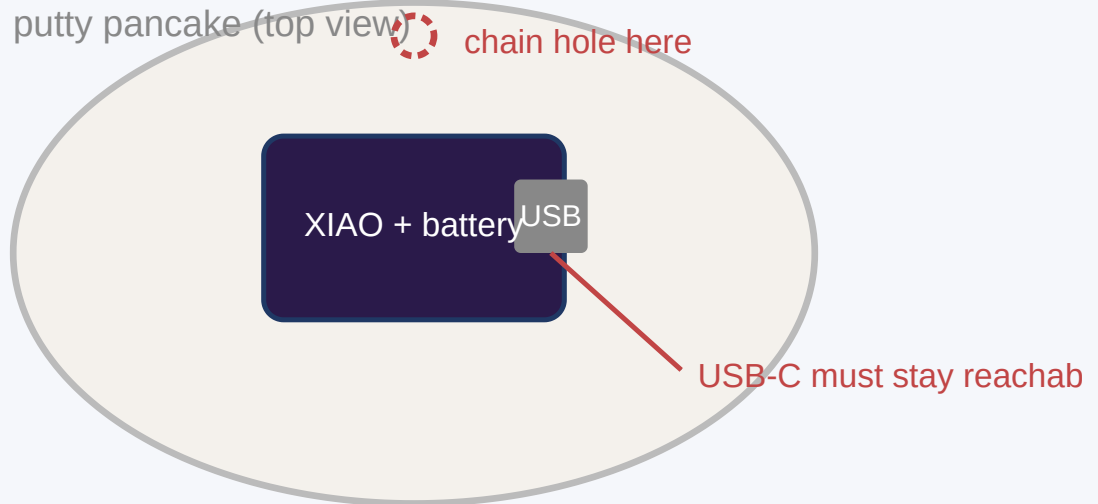
1 min

- 1 Press the ball of putty flat with your gloved palm. Aim for a pancake about **50mm long × 35mm wide × 4mm thick**.
- 2 Smooth the surface with a wet fingertip (just dip in water). This makes it look manufactured instead of bumpy.

11 Place the electronics on the pancake

2 min

- 1 Pick up your taped-together brain + battery stack. The XIAO board should be facing UP. The battery faces DOWN onto the putty.
- 2 Set it in the center of the pancake, leaving putty around all four sides.
- 3 The USB-C port should hang slightly off the bottom edge. **This is the slot you'll plug into.** Don't bury it.



Lay brain on putty pancake, leave the USB port poking out at the bottom edge.

12 Fold the putty UP around the brain

3 min

- 1 Lift the edges of the putty pancake and press them gently up the sides of the electronics.
- 2 Don't cover the top of the XIAO board completely yet. Leave a small window over the LED and the microphone.
- 3 The LED is the tiny round component near the USB end. The microphone is the small dark square on the opposite end.
- 4 Add more putty if needed to fully enclose the sides but leave the top window open.

13 Shape it into a smooth capsule

3 min

- 1 With your wet fingertip, smooth all the seams. Round the corners. Aim for a pillow-shaped capsule like the one on your website.
- 2 Use the small sculpting tools for tight spots. Slide them along edges to flatten any wrinkles.
- 3 Aim for final dimensions about **35mm × 50mm × 12mm**. Slightly bigger than your phone screen's emoji.

14 Poke the holes BEFORE it cures

2 min

- 1 **Chain loop hole** — at the very top of the pendant, push a toothpick all the way through, side to side. Make the hole about **3mm** wide. The chain will pass through here.
- 2 **LED window** — find the spot above the LED. Press a clean dot, about **3mm** wide. Don't poke all the way through, just create a thin spot so light shines through.
- 3 **Mic pinhole** — find the mic spot. Use a toothpick to poke a tiny clean hole, **1mm** wide, all the way through to the mic.
- 4 **USB-C slot** — at the bottom, make sure the USB port is still accessible. Carve away any putty covering it with the hobby knife.

 **Alternative for chain hole:** if your hole keeps closing up, skip it — after curing, glue a small jump ring (from the jewelry kit) to the top of the pendant with super glue. The chain goes through that ring instead.

15 Let it cure

1 hour (untouched)

- 1 Set the pendant aside on the parchment paper. Don't touch it.
- 2 After 10 minutes, it'll feel firm. After 1 hour, it's like ceramic. After 24 hours, it's bulletproof.
- 3 **For Tuesday's demo, 1 hour is enough.** Don't wait 24 hours unless you have the time.
- 4 While it cures, start your next pendant. Repeat Steps 8-14 to make 2 or 3 total. Spares are good.

3

Make It Look Like Metal

Liquid chrome marker turns white putty into brushed silver. 20 minutes per pendant.

16 Smooth and clean the cured pendant

5 min

- 1 Touch the pendant. If it's still soft, wait another 20 minutes.
- 2 Use a needle file (small flat one) to file down any bumps, lumps, or rough edges. Go gently — putty files easily.
- 3 Wipe the whole pendant with an alcohol prep pad. This removes oils so paint sticks. Let it dry 30 seconds.

17 Test the chrome marker on a scrap first

5 min

- 1 Grab the leftover bit of cured putty. This is your test piece.
- 2 Take the chrome marker with the medium tip (3mm). Shake it for 30 seconds (you'll hear a ball clicking inside).
- 3 Press the tip on scrap paper until silver liquid flows. This "primes" the pen.
- 4 Now paint your test piece. See how it dries shiny silver after about 60 seconds.



What you'll see: Wet, it looks like silver paint. As it dries, it turns into a real mirror finish. That's the magic.

18 Paint the pendant — first coat

5 min

- 1 Hold the pendant by the chain hole, or stick a toothpick in the chain hole as a handle.
- 2 Paint smooth even strokes. Cover one face fully. Use the big tip (3mm) for the flat parts, small tip (0.7mm) for the LED window edge and mic hole.
- 3 **Do NOT paint over the LED window itself** — leave that thin spot clear so light shines through. Same for the mic pinhole.
- 4 Don't paint inside the USB-C slot either.
- 5 Wait 60 seconds. Then paint the second face. Then the sides.

19 Wait 20 minutes, then second coat

25 min

- 1 Set the painted pendant on the parchment, propped on a toothpick so it doesn't touch the surface.
- 2 Wait 20 minutes for the first coat to fully dry. It should look mirror-shiny.
- 3 Paint a second coat over everything. Cover any thin spots or streaks from coat 1.
- 4 Wait another 20 minutes.

20 Turn mirror chrome into brushed metal

3 min

- 1 Your pendant looks mirror-shiny right now. The product on your website is BRUSHED, not mirror. Time to dull it down.
- 2 Take a microfiber cloth. Gently drag it across the chrome surface, **all in the same direction** (up-down only, not side-to-side and not circular).
- 3 After 20-30 passes, the mirror turns to fine parallel lines = brushed silver. Look at your website pendant — that's the finish you want.
- 4 If you go too far and it looks dull, stop. You can always paint another light coat to refresh.

✓ **The trick:** all strokes go the same way. Switching directions makes it look scuffed instead of brushed.

4

Finish the Pendant

Thread the chain. Power it up. You have a pendant. 5 minutes.

21 Thread the chain

2 min

- 1 Open the clasp on one of your 20" silver chains.
- 2 Pass one end through the chain hole at the top of the pendant.
- 3 Close the clasp.
- 4 If the hole closed up during curing — use a jump ring (small silver loop from the jewelry kit). Open it with pliers, hook through the hole, hook through the chain link, close it. Now the chain hangs.

22 Charge it up before demo day

1 hour passive

- 1 Plug the USB-C cable into the slot at the bottom of the pendant.
- 2 Plug into your Mac or a wall charger.
- 3 A small LED on the board should glow. The battery charges in about **1 hour**.
- 4 A full charge lasts a few hours of being on. For demo day, charge it the night before AND the morning of.

23 Final check — does it look right?

2 min

- 1 Hold the pendant up. Compare it to the photo on your website.
- 2 Smooth brushed silver capsule? ✓
- 3 Chain through the top? ✓
- 4 LED glows when plugged in? ✓
- 5 Mic pinhole visible? ✓
- 6 USB-C slot at bottom accessible? ✓



You built a pendant from scratch. A real prop with real working electronics inside, that looks like the website photo. Wear it. Get used to how it sits on your chest.

5

The Demo Setup

What sits on your desk on Tuesday. 10 minutes the morning of.

24 The story you're telling

read once

You're wearing the pendant. You say a sentence out loud, like a normal person would. The Mac on your desk hears you (Mac mic catches it because the investor is sitting right there). The action engine on the Mac infers what you meant. The browser opens. Something real happens.

The pendant is a real device with real electronics. The bridge from pendant-mic to Mac is something you ship later. For Tuesday, the Mac is the receiver. **The investor doesn't need to know that detail.** They see: spoken sentence → action.

25 Before they walk in

10 min

- 1 Pendant is fully charged. Hanging around your neck. LED is on (it's listening).
- 2 Mac is plugged in, not on battery. WiFi confirmed working.
- 3 Action engine is running on port 8787. Open a terminal, type `curl http://127.0.0.1:8787/status`. You should see "ok".
- 4 Mac mic is unmuted and Chrome is open to a clean blank page.
- 5 Practice your demo sentence 5 times. Say it like a normal person, not an actor.

If the engine doesn't act: "What you just saw was the pendant capturing the audio. The action layer is currently running in private alpha — happy to show you the recorded flow." Pull up a recorded video clip of the engine doing the action. Your founder reflex saved the demo.

If the LED is off: battery died. Plug it in to charge. "Even AI pendants need lunch." Move on to slides.

If the Mac mic doesn't catch you: repeat louder, or type the sentence into a text box that ingests it instead. Same effect.

You're ready.

Real pendant. Real electronics. Real action engine. Real demo.

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